Performance Testing & Edge-Case Validation

Template: Performance Testing and System Robustness	**Maximum Marks: 5 Marks**
Date: 15 March 2026	**Team ID:** FV-AI-2026-X1
Project Name: FoodVerse AI	**Track:** AI/ML & Generative AI

Performance Testing Metrics & Benchmarks:

Quantitative performance benchmarks measured across the FoodVerse AI endpoints under simulated network loads.

S. No	Endpoint / Feature	Target Response Time	Actual Response Time	Throughput	Pass / Fail
1	POST /api/food-recognition/analyze (Image + Text)	< 4.0 sec	1.82 sec (avg)	28 req/min	PASS
2	POST /api/freshness-detector/analyze (CV + LLM)	< 3.0 sec	1.45 sec (avg)	32 req/min	PASS
3	POST /api/hygiene-monitor/audit (Multi-Frame Video)	< 5.0 sec	2.10 sec (avg)	22 req/min	PASS
4	POST /api/kitchen-assistant/generate-recipe	< 2.5 sec	1.15 sec (avg)	45 req/min	PASS
5	POST /api/kitchen-assistant/ask-assistant	< 1.5 sec	0.68 sec (avg)	65 req/min	PASS
6	POST /api/meal-planner/generate-plan (Multi-Day)	< 3.0 sec	1.32 sec (avg)	35 req/min	PASS
7	GET / (Root Telemetry Health Check)	< 100 ms	12 ms (avg)	120 req/sec	PASS

Automated Edge-Case & Robustness Validation Suite (9/9 Passed):

Results of test_edge_cases.py executed against extreme conditions, corrupted data, and medical constraints.

Test #	Module Under Test	Edge-Case Condition Evaluated	Handling Mechanism	Test Outcome
1.1	Food Scanner	Tiny 32x32 low-resolution image input	Automated dimension upscale to >= 64px before model ingestion	PASSED (200 OK)
1.2	Food Scanner	Completely empty multipart request (no file or text)	Explicit validation guard raising HTTP 400 Bad Request	PASSED (400 Checked)
2.1	Freshness Detector	Overexposed / 100% white glare image	OpenCV saturation clamping preventing zero-division errors	PASSED (Score: 98)
2.2	Freshness Detector	Corrupted 0-byte image file	Content length validation rejecting corrupted payloads with 400	PASSED (400 Checked)
3.1	Hygiene Auditor	Dimly lit kitchen environment (mean brightness < 30)	Spatial frame illumination enhancement + Grade F safety alert	PASSED (Audited)
4.1	Kitchen Assistant	Extreme minimal ingredients ('ice cubes and tap water')	Culinary heuristic safety response generating safe preparation	PASSED (Hydration Guide)
4.2	Kitchen Assistant	Blank culinary query with whitespace	Query length guard rejecting empty question with 400	PASSED (400 Checked)
5.1	Meal Planner	Multi-allergen profile (peanuts, gluten, soy, dairy, nuts)	Strict clinical exclusion verification in LLM prompt & rules	PASSED (100% Excluded)
5.2	Meal Planner	Out-of-bounds user age (e.g. age = 150)	Pydantic v2 validation error triggering HTTP 422 Entity Error	PASSED (422 Checked)